

Project2_Template.c Code Overview

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Description:

This document provides an overview of the code written by Timothy M and Griffin T for Project2_Template.c. The code demonstrates several key concepts and functionalities, including the use of multiple buttons, SSD (Seven-Segment Display), ACL (Accelerometer), and a switch statement to manage different modes of operation.

State Machine (Timothy)

Timothy was responsible for implementing a state machine in the code. The state machine manages the program's behavior based on the current mode. Each mode represents a different stage in the program's execution.

Display Random Direction (Timothy)

Timothy implemented the displayRandomDirection function, which displays random directions on the SSD. This function generates random numbers to represent directions (up, down, left, or right) and updates the SSD display accordingly.

Mode 3 Code (Griffin)

Griffin wrote the code for Mode 3, where the player is expected to repeat a sequence of directions displayed on the SSD. The player's input is checked, and the code handles correct and incorrect inputs. The mode also keeps track of the remaining tries and transitions to other modes accordingly.

Key Components:

Initialization:

The code initializes various components, including ports, output states, and the random seed using the accelerometer (ACL). These functions are responsible for setting up the environment for the main program.

Button Handling:

The handle_raw_button_presses function manages button presses and debouncing. It keeps track of which buttons have been pressed (BtnC, BtnU, BtnD, BtnL, BtnR) and their states. Buttons are essential for user interaction and mode transitions.

Main Loop:

The main function, `main(void)`, runs in an infinite loop and manages the program's flow based on the current mode.

Switch Statement (Mode Management):

The switch statement (within the main loop) controls the behavior of the program based on the current mode. It handles transitions between different modes, including Mode 1, Mode 2, Mode 3, Mode 4, and Mode 5.

SSD (Seven-Segment Display):

The SSD is used to display information to the user, such as prompts, directions, and messages. It is managed by various functions that update the display as needed.

Program Flow:

The program starts by initializing ports and output states.

The random seed is set using the accelerometer data.

The main loop continuously checks for button presses and updates the program's behavior based on the current mode.

The switch statement inside the main loop manages the flow between different modes, each with specific actions and displays.

Mode 1 displays a "Ready" message and prompts the user to press BtnC to start.

Mode 2 is about watching and remembering a sequence of directions displayed on the SSD.

Mode 3 asks the user to repeat the displayed sequence and tracks the user's input and remaining tries.

If the user succeeds in Mode 3, the program transitions to Mode 4 and displays a "YES" message.

If the user fails in Mode 3, the program transitions to Mode 5 to indicate a game over.

The program can loop back to Mode 1 for another round.